

## 1 **Additional file 2. Supplementary methods**

2 Immune gene and protein profiles across blood, cerebrospinal fluid and peripheral nerve in  
3 inflammatory neuropathies

### 4 **Purpose and evidence attribution**

5 This appendix documents the statistical procedures and source attribution for the current  
6 manuscript. The study combines reanalysis of public transcriptomic resources with synthesis  
7 of published CSF protein estimates and localisation of published genetic candidates. These  
8 components have different inferential units and are identified separately in the figure and table  
9 index in Additional file 1.

10 Additional file 1 contains 49 worksheets with aggregate results, source documentation, gene  
11 coverage, donor design summaries, Fc sensitivity tests, refined annotation results and spatial  
12 coverage. Additional file 4 supplies the donor expression table used by the genetic  
13 localisation scripts, together with its data dictionary and validation records. Reconstruction  
14 requirements are described below.

### 15 **Blood cohorts and cross-cohort synthesis**

16 GSE211225 contains six acute, ten postacute and six control whole blood profiles. GSE31014  
17 contributes seven GBS and seven control leukocyte profiles. PRJNA1293757 contributes  
18 PBMC data from three untreated patients with AIDP and two controls. Cells were retained if  
19 they had 200 to 6000 detected genes, at least 500 counts and a mitochondrial count fraction  
20 below 20%. Each retained cell was assigned to the lineage with the highest mean  
21  $\log_{10}(\text{CPM} + 1)$  expression across its available marker genes. The monocyte markers were  
22 *LST1*, *TYROBP*, *FCER1G*, *CTSS*, *AIF1*, *LILRB1*, *S100A8* and *S100A9*. Competing lineage  
23 marker panels are specified in 02\_prjna1293757\_pseudobulk.py. Monocyte counts were  
24 summed by donor and transformed as  $\log_2(\text{CPM} + 0.5)$  for module analysis.

25 Seven script-defined gene modules were evaluated within each blood cohort. Cohort effects  
26 used Hedges g and exact label permutation tests of mean score differences. The supplied  
27 common.py enumerates all label partitions when their number does not exceed 250000. The  
28 acute versus control comparisons lie below this limit. Benjamini–Hochberg correction covers  
29 seven modules within each comparison. Acute versus postacute inferential results are omitted  
30 from the current manuscript because cross-phase donor independence could not be verified  
31 from the archived design. Postacute profiles were included in the original preprocessing.  
32 The cross-cohort model uses inverse variance weighting with the DerSimonian-Laird estimate  
33 of between-study variance. The reported modified Hartung-Knapp intervals use a residual  
34 variance scale bounded below by one and a t distribution with k minus one degrees of  
35 freedom. The repository script reproduces the estimates, intervals, P values and I<sup>2</sup> for all  
36 seven modules. Leave-one-cohort-out analyses retain two cohorts and assess direction, with  
37 no claim of confirmatory significance.  
38 CXCL8 recruitment and complement have positive effects in all three blood cohorts. Four  
39 other modules have positive effects in two cohorts. The LIF and OSM module is positive only  
40 in whole blood. None of the seven pooled effects reaches P below 0.05.

#### 41 **Peripheral nerve analyses**

42 GSE285983 provides sural nerve single nucleus data. The principal comparison includes 9  
43 CIDP and 11 CIAP donors, with eligible donor counts varying by cell group. Counts are  
44 summed for each donor and annotated cell group. Groups containing at least 20 nuclei per  
45 donor contribute to targeted pseudobulk comparisons. Library sizes use all genes. Targeted  
46 expression is transformed as  $\log_2(\text{CPM} + 0.5)$ .  
47 Within each cell group, genes are standardised across eligible donors from all diagnoses.  
48 Gene module scores are means of available standardised genes. The original CIDP and CIAP  
49 comparisons use the two sided Mann and Whitney U test and correction across eight modules

50 within each cell group. These original tests are unadjusted. The subsequent analyses add the  
51 demographic, composition, annotation and broader correction checks described below.  
52 Cell fractions are compared at donor level using unadjusted Mann-Whitney tests, with  
53 multiple testing correction across cell groups. Spearman correlations among five selected  
54 module scores use nine matched CIDP donors. The ten pairwise tests receive one Benjamini-  
55 Hochberg correction. The fifteen correlations with INCAT receive a separate correction. No  
56 result in either correlation family survives FDR 0.05. Nominal associations remain  
57 exploratory.  
58 The original Fc score was reproduced from the archived donor expression table, with a  
59 maximum absolute score difference below 0.000000000001. Reaggregation of all 20 case and  
60 comparator raw matrices reproduced 220 macrophage gene expression values to within the  
61 same numerical tolerance. Broader correction across 80 module contrasts gives  $q = 0.087$  for  
62 the Fc module, compared with  $q = 0.0087$  for the original eight module family.

### 63 **Exploratory Fc analysis specification**

64 The sensitivity analysis plan was written before the first successful new statistical run, after  
65 the original Fc association was already known. It is not a preregistration. An input coverage  
66 check identified *FCGR2C* as absent before new test results were generated. The available  
67 gene family therefore contains 11 tests. The original module includes *FCGR1A*, *FCGR2A*,  
68 *FCGR2B*, *FCGR3A*, *FCGR3B*, *FCGRT*, *FCER1G*, *TYROBP*, *SYK*, *LYN* and *HCK*.

### 69 **Demographic sensitivity and exact permutation**

70 All nine CIDP donors are male. CIAP includes seven male and four female donors. CIDP  
71 counts at Essen, Münster and Würzburg are five, one and three, respectively. Corresponding  
72 CIAP counts are one, six and four. Linear models use an intercept, a CIDP indicator, age and  
73 indicator variables for centre. The sex sensitivity adds a female indicator. HC3 covariance  
74 uses squared residuals divided by the square of one minus leverage inside the sandwich

estimator. Confidence limits and tests use a t distribution with residual degrees of freedom. These intervals describe adjusted mean score differences. The reported unadjusted comparisons use two sided Mann and Whitney P values. For individual genes, functional summaries and annotation comparisons, q values are obtained by applying Benjamini and Hochberg correction to these P values. The HC3 intervals quantify uncertainty in mean differences and are not confidence intervals corresponding to the rank tests. No effect in female CIDP donors is estimable from this design.

## **Permutation assumptions and omission analyses**

The exact centre permutation test preserves the number of CIDP labels within each centre. It enumerates all 1470 possible allocations and compares the absolute centre adjusted disease coefficient with the observed coefficient. Seven allocations are at least as extreme, giving  $P = 0.0047619$ . The observed allocation is included in the denominator, so no Monte Carlo correction is added. The test assumes exchangeability within centre and cannot remove unmeasured confounding. Separate donor, centre and gene omissions are correlated sensitivity checks. Their nominal P values are descriptive and are not treated as independent replications.

## **Functional component definitions**

Four exploratory summaries distinguish activating receptors *FCGR1A*, *FCGR2A* and *FCGR3A*, the inhibitory receptor *FCGR2B*, the neonatal Fc receptor *FCGRT*, and signalling components *FCER1G*, *TYROBP*, *SYK*, *LYN* and *HCK*. Correction covers these four summaries separately. *FCGR3B* remains in the original module and individual gene table but is outside the four summaries. Gene standardisation uses eligible donors from all diagnoses for these component analyses. Standardisation within CIDP and CIAP alone is a separate sensitivity analysis. Both component families were motivated by the original composite result.

## 98 **Refined annotation and composition**

99 The original broad macrophage annotation is the union of Macro1 and Macro2 from the  
100 whole nerve atlas. Refined immune annotations use Macro1 through Macro18 and additional  
101 immune categories. These labels refer to different annotation levels and must not be  
102 interchanged. Barcode joins retain all original macrophage nuclei. Thirty-two refined  
103 categories occur in the focused cohort, including reassigned immune populations and nuclei  
104 with no refined label. Fractions use the original macrophage count as denominator. Each  
105 category is tested at donor level, with correction across all 32 tests.

## 106 **Composition model**

107 A pseudocount of 0.5 is added to each donor category count. Log counts are centred within  
108 donor to form centred log ratios. Principal component analysis is performed across the 20  
109 donors without using disease labels. The first two components explain 22.7% and 20.5% of  
110 composition variation and enter the Fc model with age and centre. The estimated disease  
111 coefficient is  $-0.310$ , with a 95% interval from  $-1.038$  to  $0.419$  and  $P = 0.375$ . This model  
112 has 13 residual degrees of freedom. Composition could be a mediator, a confounder or a  
113 marker of technical variation. The model does not identify which explanation applies.

## 114 **Raw matrix annotation checks**

115 The 20 original count matrices are summed by donor under the original macrophage union,  
116 original Macro1, original Macro2, the refined macrophage union and each of the 18 refined  
117 macrophage subtypes. Library sizes contain all measured genes. Each donor requires at least  
118 20 nuclei for the annotation. Tests additionally require at least three eligible donors per  
119 diagnosis. Original Macro1 has no eligible donors. Three of four broad annotation definitions  
120 and nine of 18 individual refined subtypes are eligible. Broad annotation checks receive  
121 correction across three tests, and refined subtype tests receive a separate correction across

nine. Genes are standardised across eligible CIDP and CIAP donors within each definition, so effect magnitudes use definition specific scales.

#### **External data measurement and access audit**

All eight GSE285984 cell feature matrices contain the same 99 biological genes. None of the specified Fc module genes is present. The two CIDP and two CIAP spatial donors overlap the original nerve atlas, so even covered markers would provide internal triangulation. No spatial distance or expression association analysis was performed. The PRIDE PXD056286 inventory contains 19 instrument files, four Proteome Discoverer files and one checksum file. These are file counts, not participant counts. Its article supplement contains figure descriptions, with no donor protein abundance table. No new quantitative proteomic comparison was performed.

#### **CIDP blood candidates**

Kim and colleagues report four KBDS accessions, KSX10000018 through KSX10000021, for a single cell discovery study with two CIDP and two healthy donors. An analysable matrix was not retrieved for this revision. The larger clinical cohort must not be substituted for these single cell donor counts. EGA EGAD00001015725 is a controlled access candidate. The Theranostics T cell study states that data can be obtained from the corresponding author. No data access requests were sent. The audit distinguishes candidate datasets from datasets actually reanalysed and includes the source URLs.

#### **CSF evidence and published coefficients**

CSF evidence was assembled from selected reports rather than a systematic literature review. The quantitative synthesis included the ten proteins with coefficients and standard errors for both cohorts in Table 4B of Kmezic et al. These coefficients compared GBS with healthy controls on the log2 normalised protein expression (NPX) scale. The source models adjusted for age, sex and the CSF sample handling marker CCL19. All paired estimates within this

146 reported subset were included, irrespective of direction or statistical significance. Inverse  
 147 variance fixed effect estimates and normal approximation 95% confidence intervals were  
 148 calculated, with Benjamini and Hochberg correction across ten pooled tests. DerSimonian and  
 149 Laird random effects estimates,  $Q$  and  $I^2$  were also reported. The normal approximation  
 150 differs from the source regressions' small sample tests. The reported subset does not represent  
 151 the full assayed proteome. Published identifications and annotations from PXD002911 and  
 152 three other reports were recorded in a descriptive evidence map. Counts of reports and assay  
 153 types describe the coverage of these four studies. No participant level CSF matrix was  
 154 reconstructed or reanalysed.

155 Figure 3C counts unique studies and assay types within each displayed programme. Total  
 156 CSF protein concentrations are supported by the Bradford measurements reported by  
 157 Ziganshin et al. and the routine clinical concentrations reported by Sun et al. Concentration  
 158 measurements are grouped as one broad assay type, with specific methods retained in the  
 159 source table. The identification of 1402 proteins in PXD002911 does not demonstrate  
 160 increased concentration. Other assay workflows retain their published descriptions. This is a  
 161 descriptive tally of selected studies. The Kmezc coefficient synthesis is shown separately in  
 162 panels A and B.

163 IL8 is reported separately in source Table 3 and was not included in the non-IL8 Table 4B  
 164 synthesis. Its source-study findings provide context only. Nine proteins had positive  
 165 coefficients in both cohorts and  $q < 0.05$  under the fixed effect model. CD5 showed opposite  
 166 directions and pooled  $q = 0.984$ . Under the DerSimonian and Laird random effects model, the  
 167 95% intervals for IL2RA, NRP1 and IL6 included zero. Pooled normal-approximation tests  
 168 must not be substituted for the original cohort-specific regression tests.

## **Genetic candidate localisation**

The selection of *CDH4*, *DIRAS1*, *GNG7* and *SLC39A3* is based on published GWAS, Mendelian randomisation and colocalisation results from Du et al. 2024. Those estimates are retained with their source DOI. No independent GWAS, MAGMA analysis, TWAS or new colocalisation analysis was performed for this release.

Candidate localisation and the 5000-resample bootstrap were calculated from donor and cell-group summaries, using the fixed random seed 20260904. The public workbook retains disease-group expression and detection estimates, bootstrap intervals and CIDP versus CIAP contrasts. CIDP versus CIAP P values were adjusted across the 11 genes in the archived screen within each cell group. The full table contains 165 tests across 15 cell groups. Figure 5D displays four CIDP candidates in ten cell groups, and none of these forty contrasts survives FDR 0.05. The underlying Genetic\_donor\_celltype.csv is supplied in Additional file 4. The compilation and bootstrap scripts start from that table and do not reconstruct it from raw matrices. The README in Additional file 4 specifies the destination directory and commands.

## **Published external nerve observations**

Stascheit et al. reported C5b-9 deposition in 52 of 55 CIDP nerve biopsies, together with histological and proteomic observations associated with PXD056286. This study calculates the proportion and Wilson interval from that published count. It does not independently examine those biopsies or reanalyse the raw proteome. Supplementary Figure 2 has been redrawn from the retained aggregate values, with source attribution.

## **Reproduction and figure provenance**

Version 41 retains the corrected ten protein CSF synthesis and all five main figures from version 39, with the prose improvements from version 40. It adds Fc sensitivity tests, raw nerve reaggregation and source feasibility audits. Supplementary Figures 3 and 4 are new and

are generated from the retained aggregate result tables. Earlier blood and genetic matrix processing was not repeated. These statements describe the archived analysis history. The present figure revisions and the supplementary donor input are documented separately below. Figure 4A was redrawn from the archived donor mean expression summaries. All points have equal area, and colour represents the arithmetic mean of donor  $\log_2(\text{CPM} + 0.5)$  values across nine CIDP donors in each displayed cell group. The granulocyte group was omitted because matching summaries were unavailable. Figure 5B uses the arithmetic mean of donor percentages of expressing nuclei for point area and the arithmetic mean of donor mean  $\log_2(\text{CP10k} + 1)$  values for colour, with equal weight for each of the nine CIDP donors. Figure 5A identifies evidence types in individual cells. Its nerve interferon entries use cluster marker results from Supplementary Data 3 and polyneuropathy versus source control results from Supplementary Data 9 of Heming et al. The map does not rank these evidence types on a common scale. Supplementary Figure 1 retains the archived donor bootstrap intervals. Supplementary Figure 2 presents the published complement deposition count and separately describes the smaller proteomic subset. C6 protein was elevated in only two of nine CIDP cases in that source study. Original matrix-processing scripts remain available after downloading the required inputs. Recorded dependency specifications for the earlier pipeline and summary verification are retained with the v2.1.0 archive. Input\_files and Data\_reconstruction in Additional file 1 distinguish public inputs, extracted published results and locally generated intermediate tables.

## **Data and code access**

Repository release: <https://github.com/applepie567/gbs-cidp-blood-nerve-barrier/releases/tag/v2.3.1>

Version 2.3.1 contains the revised manuscript, source workbook, Figures 3 to 5, supplementary figure correction and plotting inputs. The version series is archived at <https://doi.org/10.5281/zenodo.22226672>. The GitHub release identifies the corresponding version-specific Zenodo record. Earlier versions remain available in the same series. The public repository redistributes aggregate results. The separate donor input in Additional file 4 accompanies the manuscript and is not included in the public aggregate archive. Code is provided under the MIT license. The project-derived data tables and figures are covered by LICENSE-DATA. External source materials remain subject to their original terms and attribution requirements.

## **Original files and local reconstruction**

For GSE285983, download GSE285983\_RAW.tar and GSE285983\_metadata\_all.csv.gz using the source URLs in Input\_files. In the version 2.3.1 package, place the extracted count matrices in analysis\_update/data/raw/GSE285983/h5/ and the compressed metadata file in analysis\_update/data/raw/GSE285983/. Run analysis\_update/analysis/01\_cidp\_nerve\_pseudobulk.py. All 37 original donor matrices are required to reconstruct the standardisation across diagnoses. The separate annotation reaggregation uses the 20 CIDP and CIAP matrices. The pseudobulk script writes targeted expression, module scores and cell fractions under analysis\_update/results/. The five selected CIDP module scores can then be pivoted into a donor by module table for the ten correlation tests.

Genetic localisation uses nucleus level  $\log_2(\text{CP10k} + 1)$  expression summarised by donor and cell group, with detection percentages. To reproduce the downstream genetic summaries and bootstrap, copy Genetic\_donor\_celltype.csv from Additional file 4 into analysis\_update/source\_data in the version 2.3.1 repository, then run analysis\_update/analysis/05\_compile\_genetic\_cell\_localization.py and

243 analysis\_update/analysis/08\_extended\_validation.py from the repository root. The supplied  
244 scripts do not rebuild this donor table from the original matrices. The public workbook  
245 contains the resulting group summaries, contrasts and bootstrap intervals.  
246 The donor worksheets excluded from the public aggregate workbook remain available for  
247 private author use. Missing original acquisition dates from the preceding archive have not  
248 been reconstructed. Files newly downloaded on 7 September 2026 have a separate manifest  
249 giving their actual acquisition date, source URL, byte count and SHA256 checksum. A source  
250 access check is not labelled as a file download.

### 251 **Revision-specific reproduction**

252 The retained analysis\_update directory reproduces the corrected CSF calculations and earlier  
253 figures. The strengthening\_v41 directory contains strengthen\_fc.py, subtype\_analysis.py and  
254 plot\_strengthening.py, together with aggregate results, figure files, the exploratory analysis  
255 specification and source manifests. The private checkpoint can reproduce the refined analyses  
256 with the reuse checkpoint option. Full reconstruction requires the original matrices and  
257 metadata listed in the manifest. The original donor expression table is passed explicitly as the  
258 original results directory. No new genetic bootstrap analysis was run. Run python  
259 analysis\_update/analysis/run\_v231\_figures.py to regenerate the current Figures 3 to 5 from  
260 the aggregate inputs. Historical figure runners now call the revised Figure 4 and 5 functions.